

Connected D-Hop Dominating Sets in Mobile Ad Hoc Networks

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Abstract—A mobile ad hoc network may be logically represented as a set of clusters, and the clusterheads form what is known as a dominating set in graph theory. The clusterheads can be used to perform a variety of functions for nodes in their cluster such as channel access, routing, data collection, broadcasting, etc. There have been several heuristics proposed for forming 1-hop as well as d -hop clusters. In d -hop clustering, the value of d , a constant representing the number of wireless hops in ad hoc networks, is a parameter of the heuristic and controls the size of the clusters. The problem of finding a minimum d -hop connected dominating set was shown to be NP-complete in [9], where the authors raised the question of whether the problem is still NP-complete for the restricted model of unit disk graphs. In this paper, we provide an affirmative answer to this open question. In fact, we show that the minimum d -hop connected dominating set problem is NP-complete for planar unit disk graphs with maximum degree 4. We also review some known 1-hop heuristics that we generalize to the d -hop case and compare their performance. The experimental results show that the algorithm proposed in [9] is the most efficient one.

Keywords: cluster, mobile ad hoc network, dominating set, NP-completeness

I. INTRODUCTION

By building a wireless backbone architecture in mobile ad hoc network, the communication among nodes in a wireless ad hoc network can be efficiently supported [1], [2], [11], [12]. While nodes are identical in their capabilities, certain nodes are elected to form the backbone. These nodes are clusterheads and gateways. Clusterheads are nodes that define the identity of their responsible cluster as described in [1]. Cluster topology and data are exchanged among clusters via clusterheads. Cluster members use clusterheads for routing data to other nodes in the network. Additionally, a clusterhead may also perform other control functions for cluster members such as being an arbiter for channel access of its own cluster. To achieve that, each clusterhead becomes the gateway of its cluster and clusterheads are used as the backbone of the network.

The election of nodes to be clusterheads has been a research topic in several papers (cf. [1], [2], [12], [15], [10]). Baker and Ephremides [1] devised one of the earliest clustering algorithms for ad hoc networks. This algorithm is also involved with the TDMA to avoid collision and chooses a node based on the highest ID. In a different fashion, Guha and Khuller [15]

developed a two stage algorithm to select 1-hop clusterheads using the greedy approach. Stage 1 is for choosing a node with the most uncovered neighbors, and stage 2 is for connecting clusterheads together. Ruan et. al. The most recent algorithm was given by [14] to compute connected dominating sets. The greedy approach used in this algorithm is based on two criteria: the reduction of the number of uncovered neighbors in the network and the connectedness of the clusterheads when forming clusters. In all of these papers the node¹ election guarantees that no other nodes will be more than one hop away from a clusterhead. We consider several prominent algorithms for selecting clusterheads, and compare them with the algorithm in [9]. To perform our comparison, we generalize the clustering heuristics so that a node is either a clusterhead or at most d hops away from a clusterhead, where d is a fixed parameter ≥ 1 . This paper makes the following two major contributions:

- 1) An NP-completeness proof of the minimum connected d -hop dominating set problem for unit disk graphs. This answers an open question raised in [9]. In fact, the problem remains NP-complete even when the unit disk graphs are planar and have a maximum degree ≤ 4 . This proof is fairly technical and as the reader notices, it cannot be generalized from 1-hop case to the d -hop case, $d \geq 1$, especially when the graphs under consideration are unit disk graphs.
- 2) A performance comparison of several heuristics which are generalized from 1-hop clustering to d -hop clustering. It turns out that the simple algorithm in [9] outperforms both Guha-Khuller's as well as the heuristic in [14].

The rest of this paper is organized as follows. Section II provides definitions and explanations for the proof and the heuristics. Section III shows that the problem of constructing a minimum connected d -hop dominating set is NP-complete for unit disk graphs. Several heuristics are described in Section IV, where we briefly explain the detail of heuristics that are used for the simulations in this paper. Section V shows the results

¹Throughout the remainder of this paper the terms "dominating set nodes" and "clusterheads", or "nodes" and "vertices" are used interchangeable and are synonymous

of these heuristics, and Section VI contains some concluding remarks.

II. PRELIMINARIES

In this section, we provide the definitions and explanation for some of the concepts that we use in this paper. We also describe our samples of networks that are used in the Heuristic section.

A. Network Model

In ad hoc network, a graph $G = (V, E)$ represents a network where each vertex (node) is a radio station and each edge is a communication link between two nodes. If two nodes are in the broadcast range of each other, the communication links of these two nodes are represented by two directed edges from one node to the other. In this case, an undirected edge, which is the combination of two directed edges of two nodes, is included in E . In this model, we only consider the undirected edge between two vertices (nodes). There are few restricted graph models in ad hoc network. In the scope of this paper, we only consider unit disk graph model. For graph $G = (V, E)$, this graph is connected and a set S is the subset of nodes from V of G in which all other nodes outside of S are at most d -hop away from a node in S . Set S is called the d -hop dominating set. Moreover, nodes in S are connected.

B. Clusters and Dominating Sets

Clusterheads are responsible for receiving and delivering data inside of its cluster or to another cluster. Clusterheads may be connected or not connected. If the network of clusterheads is not connected, each cluster has to use some nodes as a gateway to connect to the next cluster. A clusterhead receives the data from nodes of its responsible cluster and delivers it to the right gateway nodes depending on the stored network topology. In this paper, our attention is to the connectedness of clusterheads. Clusterheads are used as the gateway of each cluster and as the backbone of a network. There are some heuristics to select connected clusterheads. In each cluster, other nodes deliver message to a clusterhead which is at most d -hop away. The scope of this restricts to dominating sets theoretical and practical in ad hoc network only. The channel access such as TDMA and CDMA inter-cluster are not mentioned.

C. Experimental Model

The model used in assumption that all ranges are the same. The communication of nodes in the plane in the networks does not have noise or obstacles. This model for ad hoc networks is unit disk graph. An unit disk graph is defined as a set of points $p_1, \dots, p_n$ on the plane where each point is specified by their x and y coordinates and has the same transmission radius r . These points $p_1, \dots, p_n$ are generated and distributed randomly with a fix value of r . For the experimental results, we performed the heuristics on the same graph, and for each network size, we average of results of each heuristic based on 50 different graphs generated.

III. NP COMPLETENESS OF CONNECTED D-HOP DOMINATING SETS FOR UNIT DISK GRAPHS

An ad hoc network can be modeled as a graph $G = (V, E)$ and unit disk graphs are perhaps the simplest model. A set S of nodes in $G = (V, E)$ is called a d -hop dominating set if every node in V is at most d hops away from a vertex in S , where $d \geq 1$ is fixed. *Minimum connected d -hop dominating set* is the problem of determining for a graph G and an integer $k \geq 0$ whether G has a connected dominating set of size $\leq k$. In this section, we show that the minimum connected d -hop dominating set problem is NP-complete for unit disk graphs.

Theorem: Minimum connected d -hops dominating set is NP-complete for unit disk graphs.

Proof: The minimum connected d -hop dominating set problem is obviously in NP since in polynomial time, it can be checked if (1) the nodes in S are connected, (2) the size of S is $\leq k$, and (3) every node in V is either in S or at most d hops away from a node in S .

To show that the minimum connected d -hop dominating set problem for unit disk graphs is NP-hard, we construct a reduction from the Connected Vertex Cover problem for planar graphs with maximum degree 4, which was proven NP-complete in [4]. Given an instance $\langle G = (V, E), k \rangle$ of the connected vertex cover problem for planar graph with maximum degree 4, we construct an instance $\langle G' = (V', E'), k' \rangle$, of the minimum connected dominating set problem for unit disk graphs such that G has a connected vertex cover S with $|S| \leq k$ if and only if G' has a d -hop connected dominating set S' with $|S'| \leq k'$. First we use the following result by Valiant [7] to embed a planar graph with maximum degree 4 into the Euclidian plane:

A planar graph with maximum degree 4 can be embedded in the plane using $O(|V|)$ area in such a way that its vertices are at integer coordinates and its edges are drawn so that they are made up of line segments of form $x=i$ or $y=j$, for integers i and j .

Moreover, this embedding process can be done in such a way that the line segments connecting two original vertices in G must have a total length of at least 2 and the parallel line segments in G' must have a Euclidian distance of at least $(2 * d + 1)$. The set of line segments in G' representing an edge in G is called an *original edge*.

We define $\delta = (1/d)$ to be the radius of each node in the unit disk graph G' . We add nodes on the plane to V' as follow:

- 1) All original nodes of V belong to V' . These nodes are called *original nodes*.
- 2) At each coordinate point on the line segments connecting two *original nodes* u and v , we add one node, and on the line perpendicular to this line segment at this coordinate point we add successively d nodes such that the distance between adjacent nodes is δ . Nodes added at the coordinate points are called *intermediate nodes* and nodes added on the perpendicular line are called *extra nodes*.

- 3) On each unit on the line segments connecting two original nodes u and v , we add $d - 1$ nodes such that the distance between adjacent nodes is δ . The nodes added between two *intermediate nodes* are called *inside auxiliary nodes*, and the nodes added between an *original node* and an *intermediate node*, are called *outside auxiliary nodes*. At each *inside auxiliary node*, we also add d *extra nodes* on the line perpendicular to the line segment such that the distance between adjacent nodes is exactly δ .

An example network of the unit disk graph G' is shown in Figure 1 below.

In Figure 1 nodes x, y, u and v are the *original nodes*. A, B, C, D, E, and F are *intermediate nodes*. White nodes are *auxiliary nodes*. The white nodes between $(x,A), (B,y), (y,C), (C,u), (u,D)$, and (F,v) are *outside auxiliary nodes*. The white nodes between (A,B) and (D,E,F) are *inside auxiliary nodes*. Nodes added above *intermediate* and *inside auxiliary nodes* are *extra nodes*.

Obviously, the resulting graph $G' = (V', E')$ is a connected unit disk graph with radius $\delta = (1/d)$, and it can be constructed from G in polynomial time.

Claim. G has a connected vertex cover S of size

$$|S| \leq k$$

if and only if G' has a connected d -hop dominating set S' of size

$$|S'| \leq k' := k + \sum_{\{u,v\} \in E} (l_{u,v} - 1) + \left((d-1) \sum_{\{u,v\} \in E} (l_{u,v} - 2) \right) + |E|(d-1) + (k-1)(d-1)$$

where $l_{u,v}$ is the length of the edge (u,v) in G' .

Proof of Claim: Suppose that G has a connected vertex cover set S of size $\leq k$ in G . Let T be a spanning tree of S . We can construct a d -hop connected dominating set S' in G' as follow:

- 1) If $u \in V$ and $u \in S$, add the *original node* $u \in V'$ to S' .
- 2) Add all *intermediate nodes* and *inside auxiliary nodes* of V' to S' .
- 3) If $(u,v) \in T$, add the two sets of *outside auxiliary nodes* on $(u,v) \in E'$ to S' .
- 4) If $(u,v) \notin T$ and $u \in S$ and $v \in S$, add one set of neighboring *outside auxiliary nodes* from either u or v to S' .
- 5) If $(u,v) \notin T$ and $u \in S$ and $v \notin S$, add one set of neighboring *outside auxiliary nodes* of u to S' .

The total number of nodes added by rule 1 is $|S|$, by rule 2 is $(\sum_{\{u,v\} \in E} (l_{u,v} - 1)) + ((d-1) \sum_{\{u,v\} \in E} (l_{u,v} - 2))$, and by rule 3, 4 and 5 is $|E|(d-1) + (|S|-1)(d-1)$. The result of rule 3, 4 and 5 comes from the fact that at least one from two sets of *outside auxiliary nodes* on each line segments of E' representing an original edge in E has to be in S' . Moreover,

if both $u \in S$ and $v \in S$, and the edge $(u,v) \in T$, which is the spanning tree of S , then the other set of *outside auxiliary nodes* on $(u,v) \in E'$ has to be in S' as well.

The total number of nodes in the connected dominating set S' is:

$$|S'| = |S| + \sum_{\{u,v\} \in E} (l_{u,v} - 1) + \left((d-1) \sum_{\{u,v\} \in E} (l_{u,v} - 2) \right) + |E|(d-1) + (|S|-1)(d-1) \leq k'$$

Now, we can verify that S' is a connected dominating set of G' . From the rules above, all the *intermediate nodes* and *inside auxiliary nodes* are in S' . The *extra nodes* are dominated by the *intermediate nodes* and *inside auxiliary nodes*. If an *original node* v of V' is not in S' , then v and all the *outside auxiliary nodes* around v are at most d hops away from an *intermediate node*. Moreover, for every set of line segments in E' there are at least one *original node* and one set of *outside auxiliary nodes* in S' to connect with the *intermediate nodes* and the *inside auxiliary nodes*. By including all the *outside auxiliary nodes* on the line segments representing the edges of the spanning tree T of S , The nodes in S' are connected since S is a connected vertex cover set. This concludes that S' is a dominating set in G' .

Conversely, assume that we have a d -hop connected dominating set S' of size $\leq k'$. We can construct a connected vertex cover set S of size $|S| \leq k$ in G as follows.

Observe that:

- All *intermediate nodes* and *inside auxiliary nodes* have to be in S' to dominate the *extra nodes* (note that the *extra nodes* cannot be in S' , otherwise S' would not be minimum).
- For each group of line segments connecting two *original nodes* $\{u,v\} \subset V'$, at least one of $\{u,v\}$ and its neighboring *outside auxiliary nodes* have to be in S' to ensure that the *intermediate* and *inside auxiliary nodes* on these line segments are connected.
- The minimum number of original edges formed by all *original nodes* in S' is $(|N| - 1)$.

We can find a connected vertex cover set S from S' as follow:

- If $u \in V'$ is an *original node* and $u \in S'$, add u to S .
- eliminate all other nodes in S' .

In the process of forming the connected vertex cover set S , We eliminate a total of $\sum_{\{u,v\} \in E} (l_{u,v} - 1)$ *intermediate nodes* and $(d-1) \sum_{\{u,v\} \in E} (l_{u,v} - 2)$ *inside auxiliary nodes* in S' . Also, we eliminate $|E|(d-1) + (|N|-1)(d-1)$ *outside auxiliary nodes* in S' . This comes from the fact that a set of connected line segments in G' representing an original edge in G needs at least one from its two sets of *outside auxiliary nodes* to connect the *intermediate* and *inside auxiliary nodes*. Also, due to fact that S' is connected, the minimum number

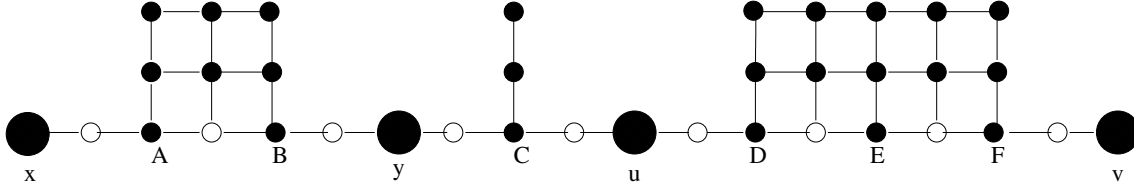


Fig. 1. Construction of $G' = (V', E')$ with $d = 2$.

of edges in E' formed by the *original nodes* in S' is $(|N|-1)$. Moreover, these *original nodes* in S' must be connected via *intermediate nodes* and *auxiliary nodes*; therefore, we eliminate the total of $(|N|-1)(d-1)$ *outside auxiliary nodes* along the connected edges of *original nodes* in S' together with $|E|(d-1)$ *outside auxiliary nodes* for all edges of E' .

Clearly, we only add the *original nodes* in S' back to S . The total number of nodes left in S' is:

$$\begin{aligned}
 |S'| &= \sum_{\{u,v\} \in E} (l_{u,v} - 1) - \left((d-1) \sum_{\{u,v\} \in E} (l_{u,v} - 2) \right) \\
 &= |E|(d-1) - (|N|-1)(d-1) = |N| \leq k' - \sum_{\{u,v\} \in E} (l_{u,v} - 1) \\
 &= \left((d-1) \sum_{\{u,v\} \in E} (l_{u,v} - 2) \right) - |E|(d-1) - (|N|-1)(d-1) \\
 &= k + (k - |N|)(d-1) \quad (*)
 \end{aligned}$$

From (*), the value of $|N|$ must be $\leq k$; therefore, $|S| = |N| \leq k$.

Finally, we can verify that S is a connected vertex cover in G . Due to the fact that all of the *intermediate* and *inside auxiliary nodes* on every edge of E' are in S' , for every original edge $(u, v) \in E$, a set of line segments in E' representing for the original edge $(u, v) \in E$ must have at least one *original node* u or v of V' to be in S' ; therefore, every edge $(u, v) \in E$ must have at least one node $u \in S$ or $v \in S$. Moreover, due to the connectedness of S' , all of the *original nodes* in S' are connected via *intermediate* and *auxiliary nodes*; therefore, by eliminating all of the *intermediate* and *auxiliary nodes* on the connected line segments in E' , which represent the original edges in E connecting these original nodes in V , the original nodes added back to S are connected. This concludes that every edge of E is covered by a node in S and the nodes in S are connected. Thus S is a connected vertex cover. This concludes the proof of the theorem.

IV. HEURISTICS

In this section, we describe the six approximation algorithms which are D -hop Connected Dominating Set (DCON), D -hop Progressive Maximum Degree (DPMD), D -hop Lowest ID Clustering (DLCA), D -hop Lowest ID of Maximum Degree (DPMD-LCA), Generalized Guha Khuller algorithm (GUHA-KHULLER), and Generalized Greedy Approximation algorithm (GREEDY-APPROX). Of those six algorithms, DCON,

DPMD and DLCA have been discussed in [9]. We combine DPMD and DLCA into DPMD-LCA, and also generalize the original GUHA-KHULLER and GREEDY-APPROX algorithms from 1-hop to d -hop. Before giving the detail of the heuristics, we note some important definitions here:

- **Black nodes:** Black nodes are clusterhead nodes.
- **Gray nodes:** Nodes that are at most d -hop away from a black node.
- **White nodes:** Nodes that are more than d -hop away from any black node.
- **d -hop neighbors:** Nodes that are at most d -hop away from a specified node.
- **Black Components:** The connected components of the sub-graph induced by black vertices.
- **White Components:** The connected components of the sub-graph induced by the nodes in G and the edges that incident to black nodes.
- **GUHA-KHULLER function:** The total number of Black Components and White vertices.
- **GREEDY-APPROX function:** The total number of Black Components and White Components.
- **Strictly connected:** Two nodes are strictly connected if they are connected by an edge.
- **Loosely connected:** Two nodes are loosely connected if they are at most d hops away from each other.
- **Loosely connected Black Component:** A set of black nodes that are loosely connected.

Let us describe the detail of all six approximation algorithms here.

DCON:

- Calculate d -hop neighbors of all nodes.
- Pick the nodes with the largest number of white d -hop neighbor, color this node black and all of its d -hop neighbors grey. Add the black node to set S .
- Consider all the 1-hop neighbors of every black node in S , calculate the d -hop neighbors for every node and color black the node with the largest white d -hop neighbors. Color its white d -hop neighbors grey. Add the black node into S and repeat this step until no white node left.
- Return the set S of black nodes as a connected dominating set.

DLCA:

- Initialize by adding all nodes into a pool.
- Continuously pick the node with the lowest ID from the pool. Color this node black and all of its d -hop neighbors

grey. Move the black node into the set S and remove grey nodes from the pool. Repeat this step until the pool is empty.

- For each node in S , calculate the shortest paths from this node to all other nodes in S . Pick two nodes with the shortest path and color black all the nodes along the path. Add these nodes into S and repeat this step until all black nodes in S are connected.
- Return the set S of black nodes as a connected dominating set.

DPMD:

- Initialize by adding all nodes into a pool.
- Continuously pick the node with the maximum d -hop neighbors from the pool. Color this node black and all of its d -hop neighbors grey. Move the black node into the set S and remove grey nodes from the pool. Repeat this step until the pool is empty.
- For each node in S , calculate the shortest paths from this node to all other nodes in S . Pick two nodes with the shortest path and color black all the nodes along the path. Adding these nodes into S and repeat this step until all black node in S are connected.
- Return the set S of black nodes as a connected dominating set.

DPMD-LCA:

- Choose a node with the lowest ID to color it black and its d -hop neighbors grey. Move this node into set S .
- Consider all the 1-hop neighbors of every node in S , pick the node with the lowest ID. Color this node black and color all of its white d -hop neighbors grey. Move this black node into S . Repeat this step until there is no white node left.
- Return the set S of black nodes as a connected dominating set.

GUHA-KHULLER:

- Initialize GUHA-KHULLER function F to be the total number of nodes.
- Consider all the white and grey nodes, Pick the one that reduces the maximum amount of GUHA-KHULLER function F using Loosely connected Black Component and color this node black. Move this node into S and repeat this step until $F = 1$ or no white node is left.
- For each node in S , calculate the shortest paths from this node to all other nodes in S . Pick two nodes with the shortest path. Color all the nodes along the path black and add them into S . Repeat this step until all nodes in S are connected.
- Return the set S of black nodes as a connected dominating set.

GREEDY-APPROX:

- Initialize GREEDY-APPROX function F = total number of nodes. And initialize $d' = d$ (number of hops).
- Consider all the white and grey nodes, Pick the one that reduces the maximum amount of F using Loosely connected Black Component. Color this node black and

all of its white d' -hop neighbors grey. Add the black node to set S and repeat this step until $F = 2$ and no white node is left.

- Repeat the following steps until $d' = 1$ and GREEDY-APPROX function $F = 2$:
 - 1) Reduce d' by 1. Calculate the actual Black Components and White Components using the Strictly connected. Reinitialize Black components and White Components with those values.
 - 2) Repeat until $F = 2$: Consider all the grey nodes. Pick one that reduces the maximum amount of GREEDY-APPROX function F using Loosely connected Black Component. Color this node black and color all of its d' -hop away neighbors grey. Add the black node into S .
 - 3) If GREEDY-APPROX function $F = 2$, then go back to step 1 above.
- Return the set S of black nodes as a connected dominating set.

We calculate the average size of dominating sets and the average running time for each heuristic on different network sizes. The result is shown in the section below.

V. EXPERIMENTAL RESULTS

To perform our experiment, we randomly generate unit disk graphs in an area of size 200x200 for sparse graphs and 100x100 for medium graphs and dense graphs. The radius used for our experimental network size of 50, 100, 150, and 200 are: 35, 25, 21, 19 for sparse graphs, 35, 25, 20, 15 for medium graphs, and 50, 40, 30, 30 for dense graphs. The average degrees of nodes are 5, 12, and 25 for sparse graphs, medium graphs, and dense graphs, respectively. For each value N (network size) and R (radius), we generate 50 different instances of unit disk graphs. For each unit disk graph generated, we run all six approximation algorithms above to find a connected dominating set. After 50 runs, we compute the average of the connected dominating set size and the average running time of each algorithm. The result are shown with the value $d = 2$ and $d = 3$ for sparse graph in figure 2, for medium graph in figure 3, and for heavy dense graph in figure 4.

The result shows that the sizes of the connected d -hop dominating sets obtained by DCON are the smallest among all six algorithms for every network size. The running time of DCON is the lowest for sparse graphs, and among the best in medium graphs and dense graphs compared with other algorithms. With regard to average running time and average dominating set size, the generalized GREEDY-APPROX is better than the generalized GUHA-KHULLER algorithm only in the case of sparse graphs, but GUHA-KHULLER algorithm seems to do better in medium graphs and dense graphs. However, both GUHA-KHULLER and GREEDY-APPROX algorithms are the best two algorithms behind DCON with regard to the average dominating set size. On the other hand, DMPD-LCA has the lowest running time. This algorithm could be useful if we want to find a large number of clusters with a small number

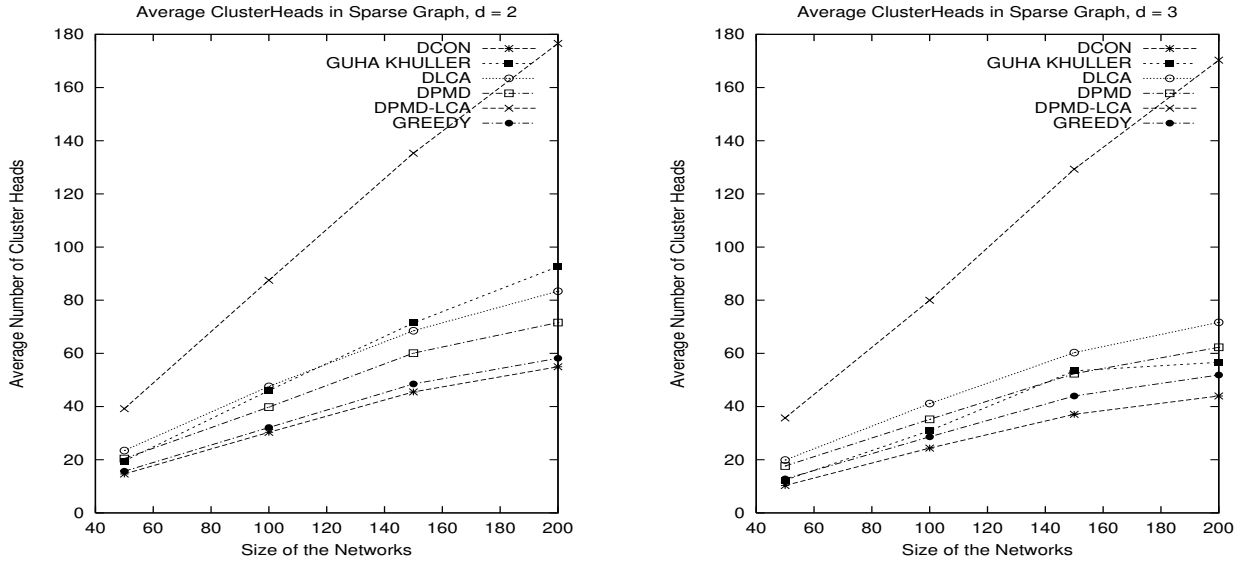


Fig. 2. Average Number of Clusterheads for Sparse Networks.

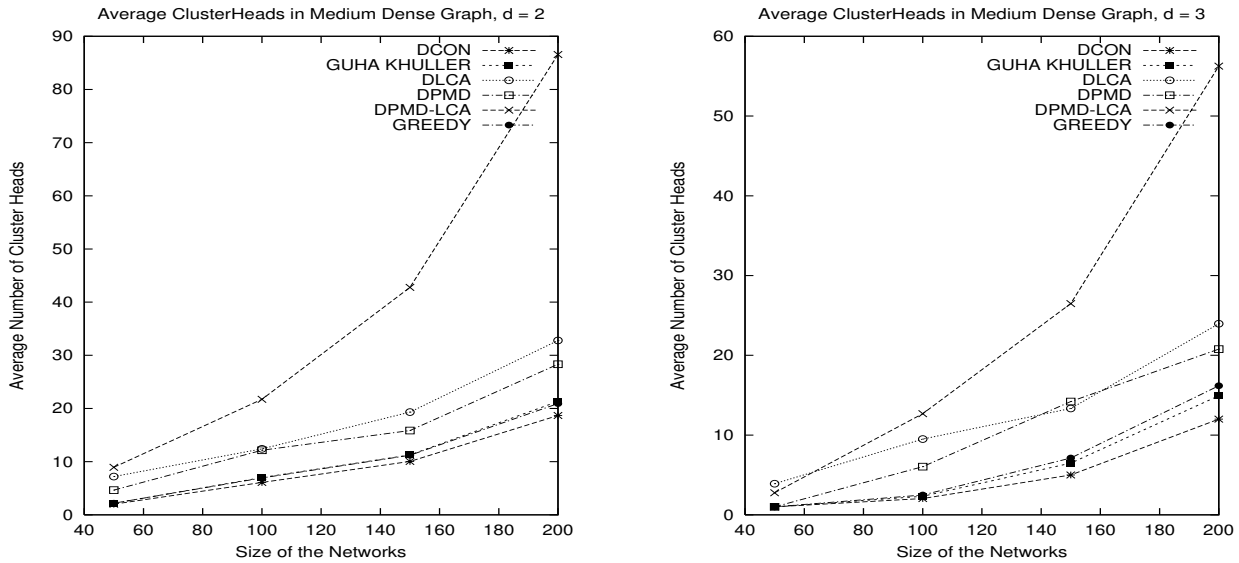


Fig. 3. Average Number of Clusterheads for Medium Networks.

of nodes in each cluster in the consideration of running time. The running times for each algorithms are shown in figure 5, figure 6, and figure 7.

VI. CONCLUSION

In this paper, we have shown that finding minimum connected d -hop dominating sets for unit disk graphs is NP-complete. Due to this result, we have to confine ourselves with good approximation algorithms. From our simulation results, the DCON heuristic for forming connected clusterheads appears to be the best among other algorithms. Moreover, the running time for DCON, compared to other heuristics, is shown to be the lowest in sparse graphs and average in

medium graphs and dense graphs. In view of these results, DCON appears to be the best heuristic in computing d -hop connected dominating sets.

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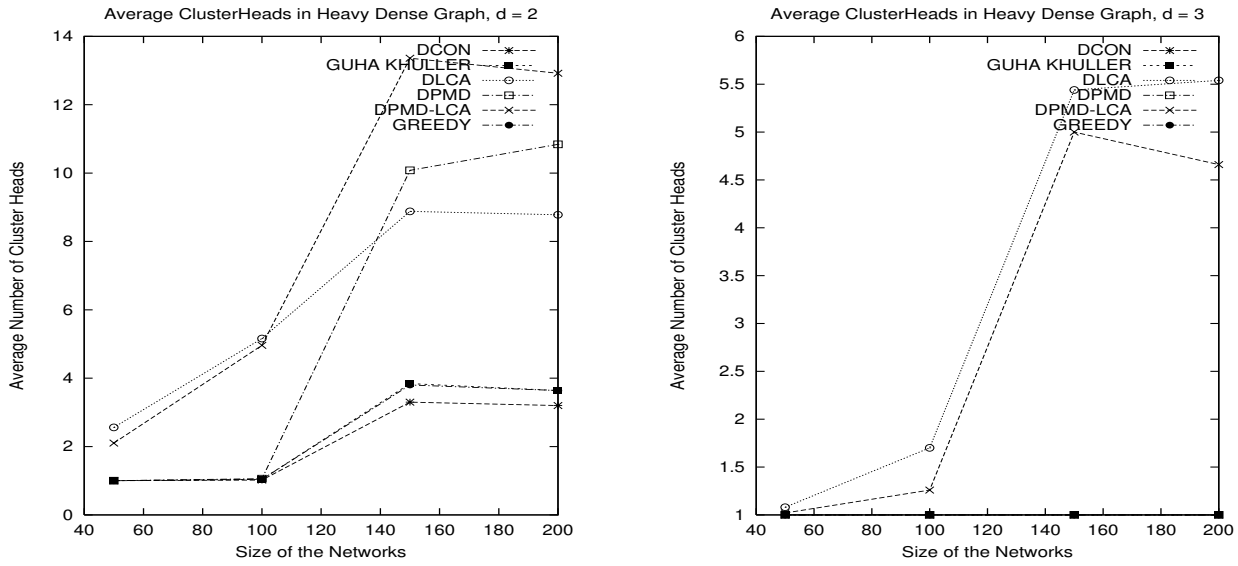


Fig. 4. Average Number of Clusterheads for Dense Networks.

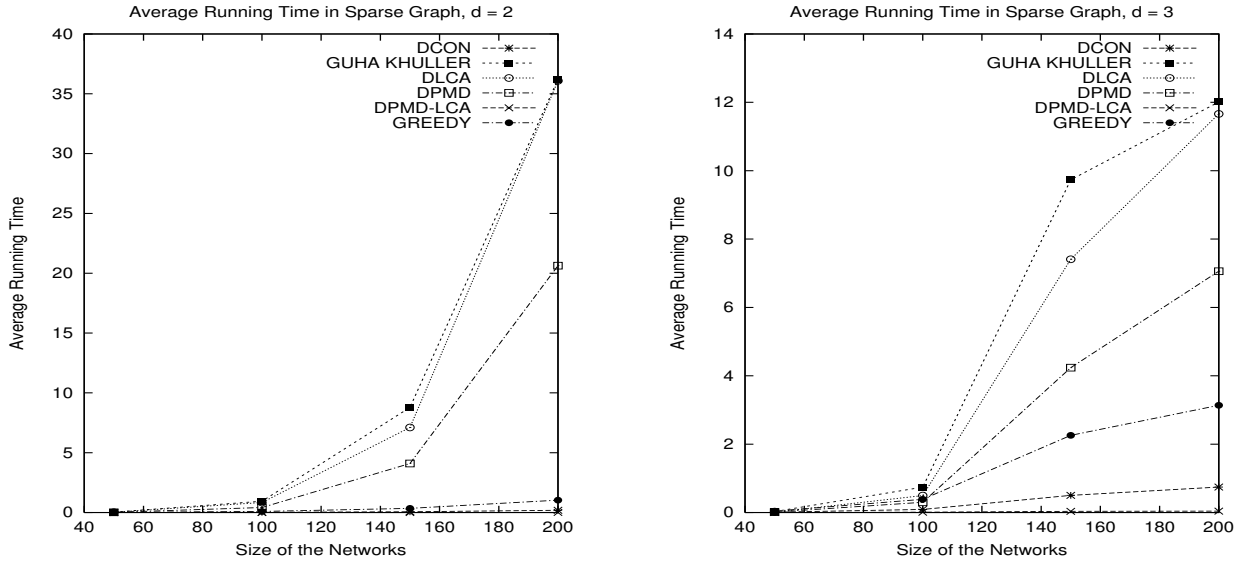


Fig. 5. Average Running Time for Sparse Networks.

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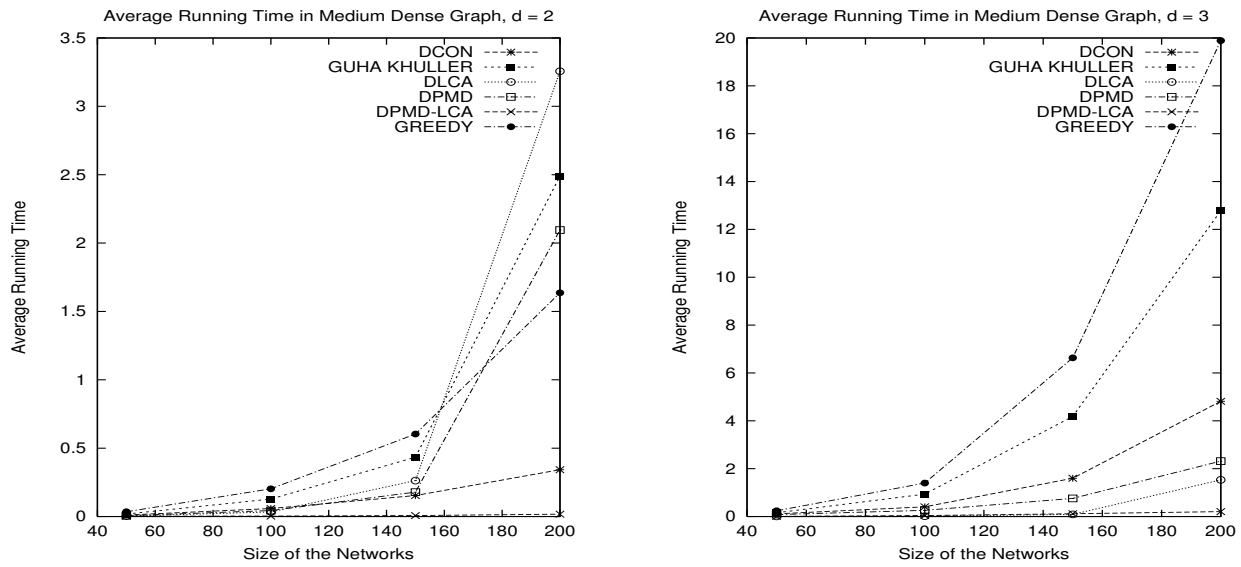


Fig. 6. Average Running Time for Medium Networks.

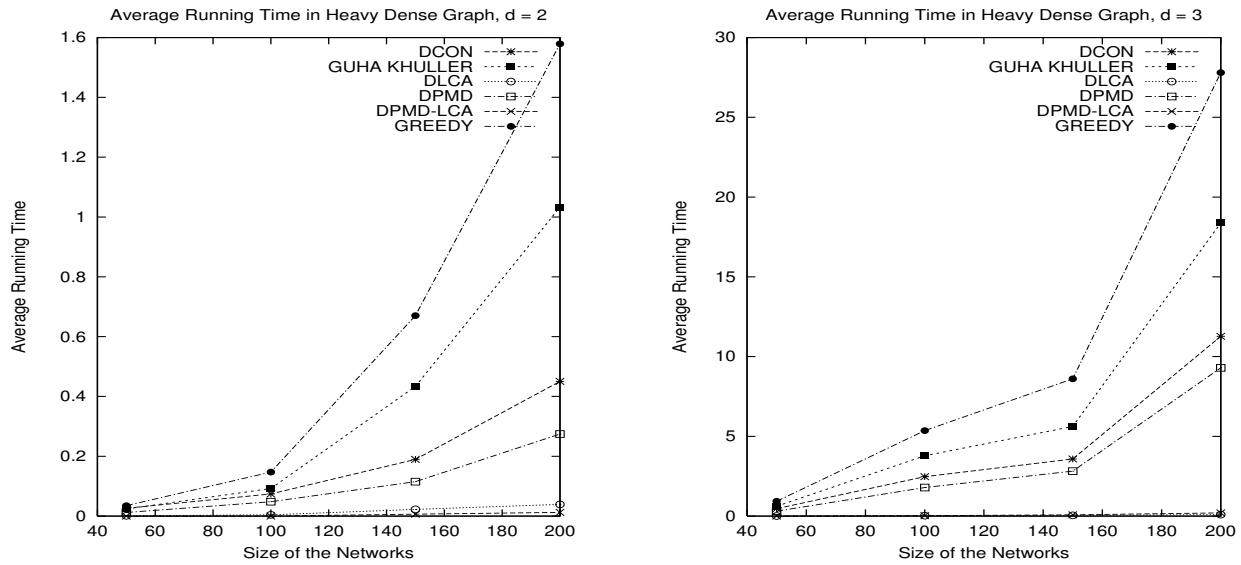


Fig. 7. Average Running Time for Dense Networks.